

Muhammad Ali Hassan

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Education

LUMS, School of Science & Engineering | B.S. Computer Science Sep 2021 – May 2025

- **Coursework:** *Distributed Systems, Blockchain, Internet of Things, Operating Systems, Network Security, Topics in LLMs, Machine Learning, Deep Learning, ICT4D, Entrepreneurship (Graduate courses mentioned in italics)*
- **Exchange Semesters:** University of Baltistan (Summer 2023)

Bloomfield Hall School | Cambridge Assessment International Education A Level Aug 2018 - May 2020

- **Grades:** Math(A*), Physics(A*), Chemistry(A*)

Research Experience

AI Research Engineer— Tibbling Technologies Jan 2025 – Present

Fourier Adaptive Transformer (FAT-1D) for Lightweight Neural Segmentation

Advisor: Dr. Asim Iqbal (Adjunct Assistant Professor, Weill Cornell Medicine)

Developed FAT-1D, a 1D segmentation model achieving 93% faster inference on CPU with 20K parameters, outperforming SAM on LiveCells/FluCells empowering real-world partners like ETH Zurich, Gubra and Broad.

- Designed a Fourier-based adaptive patching transformer architecture to prioritize attention in key frequency regions, delivering SOTA Dice scores on 36,000+ preclinical images, leading to a manuscript for *Nature Methods*.
- Validated on real-world datasets (Gubra dataset; BloodCells; LiveCell/FluCell; Boston Children's; Broad Institute), surpassing baseline Dice scores at exponentially faster inference, enabling real-time segmentation.
- Deployed a CPU-only web app of FAT-1D, empowering partners with cloud based (AWS) fine-tuning, real-time 3D-stack segmentation, and intensity analysis, facilitating rapid in-lab diagnostics without requiring a GPU.
- Performed Data Analysis for Harvard's Broad Institute on a pre-clinical dataset, extracting colocal and regional information for neuron density and count per region, directly aiding in autism drug discovery.

Research Apprentice — Cloud Computing Research Lab (CCRL), LUMS Jun 2024 - Present

Blockchain-Based AI Model Governance and Auditable Policy Compliance System

Advisors: Dr. Basit Shafiq (Associate Professor, LUMS), Dr. Jaideep Vaidya (Distinguished Professor, Rutgers)

Developed a blockchain system using BPMN workflows and LLMs for secure, decentralized policy-compliant access to AI models and healthcare datasets, addressing critical privacy concerns in highly regulated fields like healthcare.

- Developed an LLM-based workflow to generate BPMN diagrams and recommend datasets/models from natural language queries, improving workflow usability, leading to an AMIA paper on policy-compliant medical research.
- Engineered interactive workflow visualization using survival analysis models (Cox model/Kaplan-Meier curves) on SEER and German Cancer Registry datasets to enable LLM-guided survival analysis for 400,000+ patients.
- Established a Web3 pipeline for communication between smart contracts, servers, and clients using a MERN platform with INFURA APIs for interacting with the Sepolia Testnet, decentralizing policy-compliant access.
- Conducted an ablation study evaluating LLM-generated BPMN workflows across four prompt settings, achieving near-perfect validity and completeness while revealing trade-offs between prompt richness, detail, and latency.

Research Apprentice — IoT Lab, LUMS Sep 2024 – Apr 2025

Dynamic IoT Surveillance for Fire-Prone Regions

Advisor: Dr. Murtaza Taj (Associate Professor, LUMS)

Developed an IoT-based forest fire detection system using Hikvision IP cameras for real-time monitoring in Pakistan's fire-prone regions deploying across five cameras in Northern Pakistan covering 300 square kilometers.

- Optimized pan, tilt, and dynamic zoom with automated overlap calculations and hyperparameter tuning for recursive image capture, enabling flexible inspection and addressing challenges for localizing fire hotspots.
- Streamlined fire detection by integrating into existing LLMs and YOLO architecture, enabling automated hotspot localization with 99% accuracy improving early warning systems to alert authorities.
- Developed time-series GIF workflows with Python's OpenCV to automate the visualization of capture sequences, improving monitoring efficiency and disaster preparedness by authorities.

Research Intern — Centre for Water Informatics & Technology, LUMS Jun 2024 – Aug 2024

Cost-Effective AIoT Flood Monitoring System

Advisor: Dr. Talha Manzoor (Assistant Professor, LUMS)

Developed a cost-effective Edge AIoT system using ESP-32 Cam for real-time flood and wildlife monitoring in northern Pakistan, enhancing disaster preparedness and environmental protection.

- Integrated YOLOv5 with Edge Impulse and LLMs on ESP-32 Cam for real-time image analysis, achieving disaster monitoring at 1/10th the cost of traditional systems.
- Engineered GSM-based data transmission pipeline using sleep modes and watchdog timers for real-time drive uploads to ensure reliable operation in internet-limited regions with optimized power usage.
- Identified limitations in edge-AI deployment pipelines by evaluating bottlenecks across embedded systems, enabling the formulation of deployment strategies to scale AIoT cameras for reliable disaster monitoring.

Researcher – Topics in Large Language Models

Sep 2024 – Dec 2024

Identifying Bottlenecks in Deploying LLMs on Resource-Constrained Devices

Advisors: Dr. Ihsan Ayyub Qazi (Associate Professor, LUMS), Dr. Zafar Ayyub Qazi (Assistant Professor, LUMS)

Explored the deployment of Llama 3.2 (1 billion parameters, 2-bit quantized) on low-cost Android devices to optimize LLM usability for underserved users in resource-constrained environments.

- Conducted a user study highlighting the prevalence of older Android devices in Pakistan, emphasizing the need to adapt LLMs for equitable AI access akin to advanced tools like Apple Intelligence.
- Developed a memory monitoring pipeline using Ollama to host the LLM locally and the Android Debug Bridge to track metrics such as active/inactive memory, cache usage, and response latency during specific LLM prompts.
- Analyzed performance by testing different prompts across multiple iterations, identifying memory and runtime bottlenecks, highlighting the need for future benchmarking, KV-cache optimizations, and scalable AI access.

Researcher – Topics in IoT

Jan 2024 – May 2024

Security Vulnerabilities in Battery-less IoT Devices

Advisor: Dr. Naveed Anwar Bhatti (Assistant Professor, LUMS)

Investigated critical vulnerabilities in battery-less IoT devices, highlighting the need for improved security measures in resource-constrained IoT systems to mitigate risks and prevent exploitation.

- Developed a voltage monitoring pipeline using Mementos to preserve intermittent states and capacitors for non-invasive charge monitoring to track the charge/discharge cycles of the IoT device.
- Analyzed discharge patterns and device algorithms in battery-less IoT devices to uncover security flaws, revealing unique charge/discharge cycles for different apps and highlighting the need for tailored safeguards.

Teaching Experience

Teaching Assistant — School of Science & Engineering, LUMS

Aug 2022 – May 2024

CS 3812, EE 3812 – Blockchain: Technology and Applications (Spring 2023)

- Led weekly office hours and tutorials on concepts such as using Solidity and JavaScript-based contract deployment, guiding 40 graduate/undergrad students through complex concepts in blockchain.
- Enhanced course content by developing assignments, quizzes, and exam questions to ensure a challenging curriculum with 20 hours per week dedicated support.

CS 100: Computational Problem Solving (Fall 2022 & Summer 2022)

- Led teaching team as Head TA, managing 4 other TAs, lab logistics, and coordinating tutorials to establish an effective support system for 200+ students.
- Mentored C++ programming labs through control flow, functions, and data structure instructions to guide 20+ students across 6 project groups to implementation mastery.

CS 210: Discrete Mathematics (Fall 2023)

- Guided students on set theory, logic, combinatorics, graph theory, and proofs, providing assignment support and office hours to ensure understanding and strong academic performance for 200+ students.

Projects

Distributed, Fault-Tolerant Key-Value Store | Golang

Sep 2023 – Dec 2023

- Implemented a distributed key-value store using Raft consensus for leader election and log replication, ensuring data persistence, high availability, and fault tolerance across nodes during failures.

Distributed Hash Table | Python, Socket Programming

Feb 2023 – May 2023

- Developed a distributed hash table with a peer-to-peer model and fault tolerance, improving data retrieval reliability in decentralized networks.

RAG-Based Researcher Chatbot | Python, LangChain, Pinecone, FAISS

Sep 2024 – Nov 2024

- Built a research assistant chatbot using LangChain that leverages Retrieval-Augmented Generation to answer questions by retrieving relevant information from a collection of research papers and Wikipedia.

Oracle Cloud Electricity Billing System | FastAPI, Nginx, SQL, Virtual Machines

Oct 2024 – Dec 2024

- Designed and deployed a billing system web application on Oracle Cloud, enabling efficient bill retrieval, payments, and adjustments with robust error handling and a scalable three-tier architecture.

Recruitment and Onboarding | *JavaScript, MERN, Firebase, Web Sockets*

Jan 2024 – May 2024

- Built a MERN-based recruitment platform with video uploads, interactive chat, and an intuitive interface, streamlining the hiring process and improving applicant and HR engagement.

Trading App | *JavaScript, MERN, Web Sockets*

Mar 2024 – May 2024

- Engineered a real-time trading platform using the MERN stack and Socket.io, enabling instant offer updates and inventory synchronization to enhance user experience and market responsiveness.

Command-line Shell | *C*

Sep 2023 – Nov 2023

- Developed a command-line shell in C replicating UNIX shell functionality, including pipelining and command chaining, to enable efficient process management and task automation.

Research Assistant | *Python*

Jun 2022 – Aug 2022

- Identified suitable predictive models, including LSTMs and Transformers, by analyzing humidity and historical flood datasets to inform future development of improved early-warning systems for remote Pakistani regions.

Awards & Honors

- Fulbright Master's Scholarship (2026)
- Dean's Honor List (2021-2022)
- Merit Scholarship in LUMS (2021-2022)
- Winner of Startup Incubator - Ideation Academy (2023)
- Winner of Techstars Startup Weekend (2025)
- 100% Merit Scholarship in A' Levels
- Winner of Future Synergy – AI-powered Innovation Challenge for Incubation Placement, Azerbaijan Scholarship

Skills & Toolkit

Programming Languages: Python, Golang, Solidity, JavaScript, TypeScript, C, C++, Haskell, SQL, HTML/CSS

Frameworks and Libraries: PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib, MERN, LangChain, YOLO, Pinecone, OpenCV, Albumentations, FAISS

Tools and Platforms: Napari, Fiji, Docker, PM2, AWS, Azure, Hugging Face, Remix IDE, Arduino IDE, MetaMask, XACML, BPMN, Oracle Cloud, Hikvision API, Git, Linux (PuTTY, WinSCP), MongoDB

Certifications: Fundamentals of Accelerated Data Science (NVIDIA)

Leadership & Community

CEO and Co-Founder – Project Indigenous

Jan 2025 – Present

- Recognizing the scarcity of local-language AI resources, we're crafting and open-sourcing a Pashto ASR dataset, laying the groundwork for an all-Pakistan LLM that drives linguistic inclusion and equitable AI access.

Co-Founder & General Secretary – *E-Gaming at LUMS*

May 2023 – Jun 2024

- Founded and led the LUMS E-Gaming Society, hosting high-profile esports events (1000+ attendees, 200K PKR sponsorships), managing a 50+ member team, and catalyzing a thriving gaming community.

Peer Ambassador for Social Support – *LUMS*

Sep 2023 – Jun 2024

- Acted as a mentor and resource for freshmen, offering advice on academic, social, and personal matters to enhance their overall university experience, and assisting them in navigating the university's structure.

Committee Director – *LUMS Model United Nations*

May 2023 – May 2024

- Led international conference organization by directing a 50-member team in scheduling 20+ academic sessions and managing multi-venue operations to foster leadership and academic dialogue among participants.

Director Media – *SPADES & DANCELUMS*

May 2021 – May 2023

- Led university-wide media coverage by managing media teams of 10+ for PSIFI and premier events to reach an audience of 50,000+ through creative content and targeted campaigns.